

Horizontal system analog channels							
		2002A	2004A	2012A	2014A	2022A	2024A
Time base delay time range	Pre-trigger	Greater of 1 screen width or 200 μs (400 μs in interleaving mode)					
	Post-trigger	1 s to 500 s					
Channel-to-channel deskew range		± 100 ns					
Δ Time accuracy (using cursors)		± (time base accuracy ¹ reading) ± (0.0016 ¹ screen width) ± 100 ps					
Acquisition modes							
Normal							
Peak detect		Capture glitch as narrow as 500 ps at all timebase settings					
Averaging		Select from 2, 4, 8, 16, 64... to 65,536					
High resolution mode		12 bits of resolution when ≥ 20 μs/div					
Segmented		Re-arm time= 19 μs (minimum time between trigger events)					
Trigger system							
Trigger modes		• Normal (triggered): Requires trigger event for scope to trigger • Auto: Triggers automatically in absence of trigger event • Single: Triggers only once on a trigger event, press [Single] again for scope to find another trigger event, or press [Run] to trigger continuously in either Auto or Normal mode • Force: Front panel button that forces a trigger					
Trigger coupling		Coupling selections: AC, DC, noise reject, LF reject and HF reject					
Trigger source		Each analog channel, each digital channel (MSO models or DSOX2MSO upgrade, Ext, WaveGen, line)					
Trigger sensitivity (internal) ¹		< 10 mV/div: greater of 1 div or 5 mV; ≥ 10 mV/div: 0.6 div					
Trigger sensitivity (external) ¹		200 mV (DC to 100 MHz); 350 mV (100 to 200 MHz)					
External trigger input		Included on all models					
Trigger type selections							
All 2000 X-Series models							
Edge		Trigger on a rising, falling, alternating or either edge of any source					
Edge then edge (B trigger)		Arm on a selected edge, wait a specified time, then trigger on a specified count of another selected edge					
Pulse width		Trigger on a pulse on a selected channel, whose time duration is less than a value, greater than a value, or inside a time range • Minimum duration setting: 2 to 10 ns (depends on bandwidth) • Maximum duration setting: 10 s					

1. Denotes warranted specifications, all others are typical. Specifications are valid after a 30-minute warm-up period and from $\pm$ 10 °C firmware calibration temperature.
2. 1 mV/div and 2 mV/div is a magnification of 4 mV/div setting. For vertical accuracy calculations, use full scale of 32 mV for 1 mV/div and 2 mV/div sensitivity setting.



Pattern	Trigger when a specified pattern of high, low, and don't care levels on any combination of analog, digital, or trigger channels is [entered exited]. Pattern must have stabilized for a minimum of 2 ns to qualify as a valid trigger condition.
Video	Trigger on all lines or individual lines, odd/even or all fields from composite video, or broadcast standards (NTSC, PAL, SECAM, PAM-M)
Runt Trigger	on a position runt pulse that fails to exceed a high-level threshold. Trigger on a negative runt pulse that fails to exceed a low-level threshold. Trigger on either polarity runt pulse based on two threshold settings. Runt triggering can also be time-qualified (< or >) with a minimum time setting of 6~10 ns (depending on bandwidth) and maximum time setting of 10 s.
Rise/fall time	Trigger on rise-time or fall-time edge speed violations (< or >) based on user-selectable threshold. Select from (< or >) and time settings range between 3-5 ns (depending on bandwidth) and 10 s
Nth edge burst	Trigger on the Nth (1 to 65535) edge of a pulse burst. Specify idle time (10 ns to 10 s) for framing. Pattern Trigger when a specified pattern of high, low, and don't care levels on any combination of analog, digital, or trigger channels is [entered exited]. Pattern must have stabilized for a minimum of 2 ns to qualify as a valid trigger condition. Minimum duration setting: 6-10 ns (depending on bandwidth) and 10 s Or: Trigger on any selected edge across multiple analog or digital channels
I ² C (optional)	Trigger on I ² C (Inter-IC bus) serial protocol at a start/stop condition or user defined frame with address and/or data values. Also trigger on missing acknowledge, address with no ack, restart, EEPROM read, and 10-bit write.
SPI (optional)	Trigger on SPI (Serial Peripheral Interface) data pattern during a specific framing period. Supports positive and negative Chip Select framing as well as clock Idle framing and user specified number of bits per frame.
CAN (optional)	Trigger on CAN (controller area network) version 2.0A and 2.0B signals. Trigger on the start of frame (SOF) bit (standard). Remote frame ID (RTR), data frame ID (~RTR), remote or data frame ID, data frame ID and data, error frame, all errors, acknowledge error and overload frame.
LIN (optional)	Trigger on LIN (Local Interconnect Network) sync break, sync frame ID, or frame ID and data
RS232/422/485/UART (optional)	Trigger on Rx or Tx start bit, stop bit or data content
Cursors	
Types	Amplitude, time, frequency (FFT), manual, tracking, binary, HEX
Measurements	ΔT , $1/\Delta T$, $\Delta V/X$, $1/\Delta X$, ΔY , Phase and Ratio
Cursors ²	- Single cursor accuracy: $\pm$ [DC vertical gain accuracy + DC vertical offset accuracy + 0.25% full scale] - Dual cursor accuracy: $\pm$ [DC vertical gain accuracy + 0.5% full scale] ¹
Automatic waveforms measurements	
Voltage	Snapshot all, maximum, minimum, peak-to-peak, top, base, amplitude, overshoot, preshoot, average- N cycles, average-full screen, DC RMS- N cycles, DC RMS- full screen, AC RMS- N cycles, AC RMS- full screen (std dev)

